AP Calculus BC Test 5

H23 Parametric equations

A parametric equation is a pair of equations that define a curve in the plane. The equations are typically written in the form $x = f(t)$ and $y = g(t)$, where t is a **parameter** that varies over some interval.

The curve defined by the parametric equations can be traced out by varying the parameter t from its starting value to its ending value, drawing arrows to indicate directionality.

Eliminating the parameter

To eliminate the parameter, solve one of the equations for t and then substitute that expression into the other equation.

Example: Given the parametric equations $x = 2t+1$ and $y = t^2-3$, eliminate the parameter to find an equation in terms of x and y.

$$x = 2t + 1$$
$$t = \frac{x-1}{2}$$
$$y = t^2 - 3$$
$$y = \left(\frac{x-1}{2}\right)^2 - 3$$
$$\boxed{y = \frac{(x-1)^2}{4} - 3}$$

For trigonometric functions, use a Pythagorean identity.

Example: Given the parametric equations $x = 4\cos(t)$ and $y = 4\sin(t)$, eliminate the parameter to find an equation in terms of x and y. Determine the shape of the curve.

$$x = 4\cos(t)$$
$$y = 4\sin(t)$$

Using the Pythagorean identity $\sin^2(t) + \cos^2(t) = 1$:

$$\left(\frac{x}{4}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$$
$$\boxed{\frac{x^2}{16} + \frac{y^2}{16} = 1}$$

The curve is a $\boxed{\text{circle}}$.

Tangent lines

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Example: Find the slope of the tangent line to the curve and the concavity of the curve defined by the parametric equations $x = 4\cos(t)$ and $y = 4\sin(t)$ at the point where $t = \frac{\pi}{4}$.

Step 1: Find the first derivative and plug in the value of t to get the slope.

$$\begin{aligned}
\frac{dy}{dx} &= \frac{\frac{dy}{dt}}{\frac{dx}{dt}} \\
&= \frac{4\cos(t)}{-4\sin(t)} \\
&= -\cot(t) \\
-\cot(t)|_{t=\frac{\pi}{4}} &= -1 \\
&\boxed{m = 1}
\end{aligned}$$

Step 2: Find the second derivative and plug in the value of t to get the concavity.

Since we know $\frac{dy}{dx} = -\cot(t)$, differentiate and divide by $\frac{dx}{dt}$ to find the second derivative:

$$\begin{aligned}
\frac{d^2y}{dx^2} &= \frac{\frac{d}{dt}\left(-\cot(t)\right)}{\frac{dx}{dt}} \\
&= \frac{\csc^2(t)}{-4\sin(t)} \\
-\frac{\csc^2(t)}{4\sin(t)}|_{t=\frac{\pi}{4}} &= -\frac{\sqrt{2}}{2} \\
&\boxed{\text{concave down}}
\end{aligned}$$

Horizontal and vertical tangents

If $\frac{dy}{dt} = 0$ and $\frac{dx}{dt} \neq 0$, then there is a **horizontal tangent**.

If $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0$, then there is a **vertical tangent**.

Arc length formula

For parametric equations $x = f(t)$ and $y = g(t)$, the arc length from $t = a$ to $t = b$ is given by the formula:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

H24 Vectors

A vector is a quantity that has both magnitude and direction. Vectors can be represented in component form as $\mathbf{v} = \langle v_1, v_2 \rangle$.

position vector $\mathbf{r}(t) = \langle x(t), y(t) \rangle$

velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$

acceleration vector $\mathbf{a}(t) = \langle x''(t), y''(t) \rangle$

Below are the formulas for displacement, speed, and distance travelled:

Displacement (vector), the change in position (points from the starting position to the ending position)

$$\left\langle \int_{t_0}^{t_1} v_x(t)\, dt, \int_{t_0}^{t_1} v_y(t)\, dt \right\rangle$$

(note that $v_x(t)$ **and** $v_y(t)$ **are equivalent to** $x'(t)$ **and** $y'(t)$, respectively)

Speed (scalar), the magnitude of the velocity vector

$$\sqrt{v_x(t)^2 + v_y(t)^2}$$

or

$$\sqrt{x'(t)^2 + y'(t)^2}$$

Distance travelled (scalar), the total length of the path taken

Same as the arc length formula:

$$\int_{t_0}^{t_1} \sqrt{v_x(t)^2 + v_y(t)^2}\, dt$$

Example: An object moving along a curve in the xy-plane has position $(x(t), y(t))$ at time t with $\frac{dy}{dt} = 2 + \sin(e^t)$. The derivative $\frac{dy}{dx}$ is not explicitly given. At $t = 3$, the object is at the point $(4, 5)$. Find the y-coordinate of the position at time $t = 1$.

Step 1: Thinking logically, to find the y-coordinate at $t = 1$, we work backwards from $t = 3$ to $t = 1$ using the given derivative $\frac{dy}{dt}$. $y(3)$ can be found by integrating $\frac{dy}{dt}$ from $t = 1$ to $t = 3$ and adding the result to $y(1)$, so to find $y(1)$, we can use the following formula:

$$y(1) = 5 - \int_1^3 (2 + \sin(e^1))\, dt$$

The change is being subtracted, not added, as we are going earlier in time.

Solving with a calculator, $\boxed{y(1) \approx 1.27}$

H25 Polar coordinates

A polar coordinate is written in the form (r, θ), where r is the distance from the origin and θ is the angle measured counterclockwise from the positive x-axis.

To convert polar to rectangular coordinates:

$$x = r\cos(\theta)$$
$$y = r\sin(\theta)$$

Area of a polar region

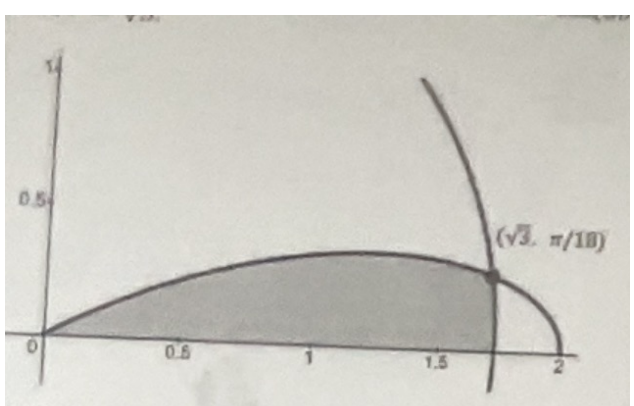
The area of a region bounded by a polar curve $r = f(\theta)$ from $\theta = a$ to $\theta = b$ is given by the formula:

$$A = \frac{1}{2}\int_a^b (f(\theta))^2\, d\theta$$

Example: Find the area bounded as shown by $r(\theta) = 2\cos(3\theta)$ and $r = \sqrt{3}$ with a calculator.

Step 1: Graph the equation.

The area is the region between the two curves, which is shaded in the graph below:



Step 2: Find the points of intersection.

The point of intersection $(\sqrt{3}, \frac{\pi}{18})$ is already given, but can be found by setting the two equations equal to each other $(2\cos(3\theta) = \sqrt{3})$ and solving for θ.

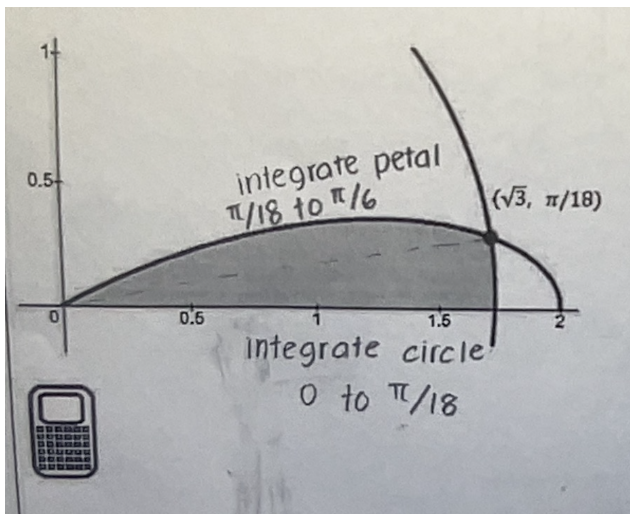
Step 3: Find where the bounds of the area are.

The bounds of the area of the circle are $\theta = 0$ to $\theta = \frac{\pi}{18}$, and the bounds of the area of the petal are $\theta = 0$ to $\theta = \frac{\pi}{6}$ (found using a calculator).

Step 4: Find the area.

When integrating with respect to θ, the bound is no longer the x-value or the y-value but the angle θ. Essentially, we are finding the area underneath the function to the angle, not the area underneath the function to an axis.

Looking at the graph, the integral of the circle wrt θ can be added to the integral of the petal wrt θ from where the circle ends $(\frac{\pi}{18})$ to the end of the petal $(\frac{\pi}{6})$ to get the total area.



First, the area of the circle that constitutes part of the petal:

$$A_{\text{circle}} = \frac{1}{2}\int_0^{\frac{\pi}{18}} (\sqrt{3})^2 \, d\theta$$

Now, the area remaining is calculated by taking the upper piece only (from $\pi/18$ to $\pi/6$):

$$A_{\text{petal}} = \frac{1}{2}\int_{\frac{\pi}{18}}^{\frac{\pi}{6}} (2\cos(3\theta))^2 \, d\theta$$

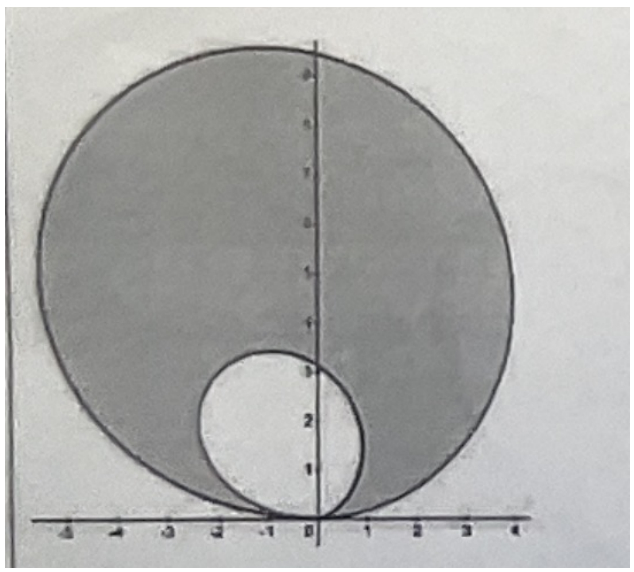
Using a calculator, computing

$$A = \frac{1}{2}\int_0^{\frac{\pi}{18}} (\sqrt{3})^2 + \ d\theta + \frac{1}{2}\int_{\frac{\pi}{18}}^{\frac{\pi}{6}} (2\cos(3\theta))^2 \, d\theta$$

$$\boxed{A \approx 0.467 \text{ units}^2}$$

Example: Find the area between the outer loop and inner loop of $r = 2\theta \sin\theta$ for θ from 0 to 2π, inclusive.

Step 1: Graph the equation.



Step 2: Find the bounds of the inner and outer loops with a calculator.

Inner: $0 \leq \theta \leq \pi$
Outer: $\pi \leq \theta \leq 2\pi$

Step 3: Subtract the outer area from the inner area. Use a calculator to solve.

$$\int_0^{\pi} \frac{1}{2}(2\theta \sin\theta)^2 \, d\theta - \int_{\pi}^{2\pi} \frac{1}{2}(2\theta \sin\theta)^2 \, d\theta$$
$$= \boxed{62.013 \text{ units}^2}$$